

Performance & Scalability Goals (Backend)

Goals:

- UI loads within 2 seconds (frontend-facing)
- API response < 400ms
- 10,000 concurrent learners
- 99% uptime target

Backend techniques to meet them:

- DB indexing on: `attempts(user_id, created_at)`, `questions(lesson_id, topic_skill(user_id, topic))`
- Caching for recommended lessons and static content
- Async workers for heavy tasks (exports, aggregation, notifications)
- Horizontal scaling behind a load balancer
- Rate-limiting AI endpoints